Lab 3: Component Modelling & Architectural Pattern Selection

Personalized Meal & Diet Subscription Manager

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Project Title	Personalized Meal & Diet Subscription Manager (PMDSM)
Domain	Health & Wellness — Personalized Nutrition / Subscription Commerce
Key Stakeholders	Subscriber, Dietitian

1. Project Scenario

The Personalized Meal & Diet Subscription Manager (PMDSM) is a subscription-based platform that automatically generates personalized weekly meal plans for subscribers based on their dietary profile, while giving a Dietitian clinical oversight to review and approve every plan before it reaches the subscriber. The system must deliver a complete subscription-to-delivery experience across the following requirements.

Core Functionality

- Subscribers can create and update a dietary profile capturing allergies, calorie goals, and diabetic limits so that meal plans are tailored to their medical needs.
- The system automatically generates a personalized weekly meal plan for each active subscriber based on their dietary profile and preferences.
- A Dietitian reviews every pending plan in a review queue and approves or rejects it with comments before it is released to the subscriber.
- Subscribers receive automated notifications when their plan is ready, approved, or rejected, as well as billing and renewal reminders tied to their subscription.
- Every plan-generation or profile-update request is tied to an active, paid subscription, which is verified before the request is processed.

Technical Constraints

- Must integrate with an external payment gateway to process subscription billing, renewals, and refunds rather than storing card data directly.
- Must deliver time-sensitive notifications (plan-ready, approval, billing) through an external email/SMS gateway.
- Must remain responsive during the weekly plan-generation cycle, when personalization requests spike sharply for every active subscriber at once.
- Should integrate with a nutrition/ingredient reference database to validate calorie counts and allergy exclusions during plan generation.

2. Architectural Style Evaluation

Three candidate architectural styles were evaluated against the scenario above before an architecture was selected.

Style	Structure	Fit for this Scenario — Pros	Fit for this Scenario — Cons
Layered	Horizontal layers: presentation, business logic, data/integration	Clean separation between the subscriber app, dietitian console, and profile/payment/notification integrations; simple to build and operate for a moderate subscriber base.	Less elastic if meal-plan generation needs to scale far beyond the steadier review and notification workload.
Microservices	Small, independently deployable services	Each capability (plan generation, dietitian review, subscription/payment, notifications) can scale and deploy independently, matching each service's very different load pattern.	Operational complexity, distributed data consistency, and network latency overhead compared to a single deployable unit.
Client-Server	Centralized server with multiple thin clients	Simple to reason about; centralized data consistency for subscriber and dietitian data.	Blurs the boundary between plan-generation logic, review workflow, and payment/notification integrations, making it harder to isolate and scale each concern.

Microservices Architecture was selected as the most appropriate style; the full justification is presented in Section 6.

3. Identified Components

Six components were identified for the Microservices Architecture, grouped by presentation, core domain, and data/integration responsibility.

#	Component	Layer	Responsibility
1	Subscriber Web/Mobile App	Presentation	Self-service UI through which subscribers set up their dietary profile, view their weekly meal plan, and track their subscription/billing status.
2	Dietitian Console	Presentation	Staff-facing UI for the Dietitian to view the review queue and approve or reject pending meal plans with comments.
3	Meal Plan Generator Service	Core Domain	Central orchestrator that fetches a subscriber's dietary profile, generates a personalized weekly meal plan, validates the subscription, and submits the plan for review.
4	Dietitian Review & Approval Service	Core Domain	Manages the review queue and the approve/reject-with-comments workflow for every generated meal plan.
5	Dietary Profile & Subscription Service	Data/Integration	Stores each subscriber's allergies, calorie goals, and diabetic limits, and manages subscription/billing state via an external payment gateway.
6	Notification Service	Data/Integration	Sends plan-ready, approval/rejection, and billing/renewal notifications through an external email/SMS gateway.

4. Component Interfaces

Eight interfaces connect the components above, each shown in the diagram using UML ball (provided) and socket (required) notation.

Interface	Provided By	Required By	Purpose
IManageProfile	Dietary Profile & Subscription Service	Subscriber Web/Mobile App	Create/update dietary profile & preferences; view subscription and billing status.
IViewMealPlan	Meal Plan Generator Service	Subscriber Web/Mobile App	Request generation of, and track, the current personalized weekly meal plan.
IFetchDietaryProfile	Dietary Profile & Subscription Service	Meal Plan Generator Service	Retrieve the subscriber's allergies, calorie goals, and diabetic limits to personalize the plan.
IValidateSubscription	Dietary Profile & Subscription Service	Meal Plan Generator Service	Confirm an active, paid subscription before a new plan is generated or released.

Interface	Provided By	Required By	Purpose
ISubmitForReview	Dietitian Review & Approval Service	Meal Plan Generator Service	Hand off a newly generated plan into the review queue and receive the approval decision.
IReviewQueue	Dietitian Review & Approval Service	Dietitian Console	Retrieve pending plans and record an approve/reject decision with comments.
ISendApprovalNotification	Notification Service	Dietitian Review & Approval Service	Trigger a subscriber notification once a plan is approved or rejected.
ISendBillingAlert	Notification Service	Dietary Profile & Subscription Service	Trigger billing, renewal, and receipt notifications tied to the subscription.

5. UML Component Diagram

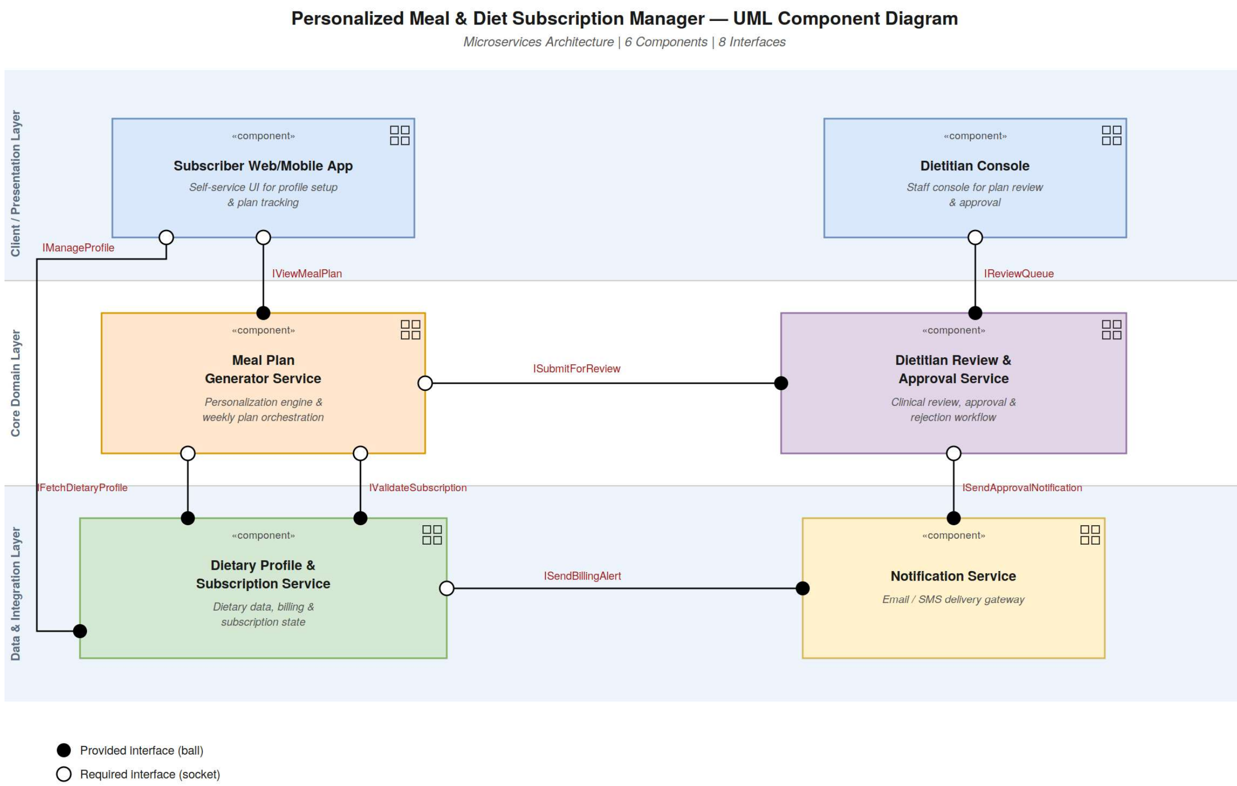

Figure 1. Component diagram for the Personalized Meal & Diet Subscription Manager (Microservices Architecture) — 6 components and 8 interfaces shown with UML ball-and-socket notation.

The editable source file (PMDSM_Component_Diagram.drawio) can be opened directly in draw.io / diagrams.net.

6. Written Architectural Justification

Architecture Selection: *We chose Microservices Architecture for the Personalized Meal & Diet Subscription Manager (PMDSM).*

Reason 1 — Independent scaling of a compute-and-schedule-driven workload

Weekly meal-plan generation runs a personalization algorithm against every active subscriber's dietary profile and fires in a sharp burst at the start of each billing/refresh cycle, while the dietitian review workflow and notification delivery run continuously at a much smaller, steadier volume. Microservices let the Meal Plan Generator Service scale out on its own for that weekly burst, without over-provisioning the review, profile, and notification services that never see that spike.

Reason 2 — Independent evolution of a human-in-the-loop workflow

The Dietitian Review & Approval Service encodes a distinct clinical business process — reviewing, and approving or rejecting plans with comments — that is likely to change on its own schedule (for example, new clinical approval rules) separate from subscription billing or plan-generation logic. A microservices split lets this workflow be updated and redeployed independently of the rest of the system.

Security Advantage

Isolating the Dietary Profile & Subscription Service as its own microservice confines sensitive medical data (allergies, diabetic limits) and payment/billing data behind one narrowly-scoped service boundary with its own access controls. A compromise of another service, such as Notification, does not automatically expose that dietary or billing data — unlike in a single shared-database monolith where all modules share one trust boundary.

Performance Benefit

Because the Meal Plan Generator Service is independently deployable, it can be horizontally scaled with extra instances purely during the weekly plan-generation window, keeping generation latency low for subscribers without paying to scale the review, profile, or notification services, which do not share that load pattern.